

Bhavana Nare

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PROFESSIONAL SUMMARY

- Senior Software Engineer with 10+ years of experience designing and deploying AI/ML-enabled platforms, including data pipelines, AI integration automation, and full stack engineering.
- Strong background in ML integration and AI platforms, combining cloud-native architecture, MLOps workflows, and DevOps automation to operationalize ML capabilities at scale.
- Proven impact architecting AI-driven systems that reduced operational risk, cut manual effort by 60–70%, and scaled into enterprise CI/CD environments.
- Expertise in system architecture, performance optimization, and automation, with hands-on leadership across the full SDLC.

EDUCATION & RESEARCH

- **Master of Computer Science (Thesis)** — University of Georgia, Athens, GA, Aug 2021 – May 2023 | GPA: 3.7 / 4.0
- **Bachelor of Technology in Computer Science** — Sree Vidyanikethan Engineering College, Tirupati, India, Oct 2010 – Apr 2014 | GPA: 7.9 / 10

PROFESSIONAL EXPERIENCE

Cybersecurity AI Analyst

Rivian Automotive (May 2025 –Present)

- Architected and deployed Python tool Beacon, a low latency AI powered risk decisioning and automated remediation system operating across 120 repositories and 400 to 600 merge requests per day.
- Designed and optimized a two-stage real time decisioning pipeline using a lightweight ML prefilter followed by LLM and RAG based deep analysis, achieving 70% noise reduction.
- Implemented production grade RAG architecture using embeddings, repository history, CODEOWNERS metadata, and historical exploit patterns to enable context aware AI risk scoring.
- Reduced inference latency to 25 seconds using asynchronous orchestration, batched embedding retrieval, and prompt context optimization.
- Delivered automated remediation workflows including merge request comments, Slack alerts, and policy citations, reducing manual triage effort by 60 to 70%.
- Integrated deterministic guardrails using Pydantic schema validation and mandatory policy citation enforcement to maintain compliance and auditability.
- Designed and enforced policy-driven validation and audit controls within CI/CD systems to ensure traceable and compliant decision workflows aligned with enterprise governance requirements.
- Built a continuous learning loop using Databricks, re indexing developer feedback into the RAG pipeline to improve classification accuracy over time.
- Led production incident triage for AI driven merge gating workflows, implementing rollback strategies and reliability controls to maintain 99% system uptime.

MLOps Engineer

Robert Bosch (Aug 2023 –April 2025)

- Designed and deployed a cloud-native risk signal processing pipeline (EventBridge → Lambda → S3 → DynamoDB) ingesting CI/CD telemetry across ~80+ repositories to generate structured behavioral signals.
- Engineered deterministic and ML-driven risk features (file-diff volatility, failure frequency, integration instability, review latency, historical defect density) to model pull-request risk propensity.
- Developed supervised classification models (logistic regression, tree-based models) for PR risk scoring, categorizing changes into Testing / Pipeline / Software risk domains.
- Implemented rule + model hybrid decision logic combining metadata heuristics with statistical scoring to improve precision and reduce false-positive alerts.
- Built online scoring services using Python to generate near-real-time risk assessments for incoming pull requests.
- Established monitoring pipelines (CloudWatch metrics, structured logging) to track model drift, feature distribution shifts, and scoring stability.
- Delivered risk dashboards visualizing throughput volatility, failure clustering, and release risk concentration, enabling engineering leadership to prioritize high-risk integrations.

- Mentored junior engineers on ML pipeline design, CI/CD automation, and distributed system reliability practices.
- Led cross-functional architecture discussions across security, DevOps, and platform teams to align AI systems with enterprise risk standards.
- Designed and implemented RESTful APIs supporting internal applications, automation services, and third-party integrations, ensuring secure data exchange and version-controlled endpoints

Senior Software Engineer –Computer Vision

Continental Automotive (May 2019 – July 2021)

- Contributed to ML-based computer vision object detection pipelines for ADAS camera systems, delivering perception outputs consumed by Emergency Braking Assistance (EBA) and other safety-critical downstream functions.
- Integrated and curated labeled training datasets into YOLO-based object detection pipelines, enabling iterative model training, validation, and performance evaluation across multiple data versions and operating conditions.
- Supported the transition from 2D to 3D bounding box detection, improving object localization, depth awareness, and spatial accuracy for perception outputs used in ADAS decision-making logic.
- Collaborated cross-functionally with labeling, software engineering teams to analyze detection behavior, identify failure modes, and refine model outputs against functional requirements.
- Validated perception outputs against accuracy, recall/precision thresholds, and production-readiness criteria, ensuring alignment with ADAS safety and performance expectations.
- Assisted in evaluating model performance across diverse scenarios and edge cases, supporting data-driven decisions for model iteration and deployment readiness.
- Built Python-based data processing pipelines to normalize detection outputs and generate longitudinal performance insights for model regression monitoring.

ADDITIONAL EXPERIENCE

- **Python Developer & Data Analyst** — Teradata India Pvt. Ltd August 2018 – May 2019
- **Senior Software Engineer** — Tata Consultancy Services (TCS) June 2014 – August 2018

TECHNICAL SKILLS

- **Programming & Languages:** Python, Java, C++, SQL, Bash/Shell, JavaScript (React)
- **AI / ML & LLM Platforms:** Vertex AI, AWS Bedrock, PyTorch, scikit-learn, Pandas, NumPy, OpenCV
- **Cloud & MLOps:** AWS (Lambda, SageMaker, S3, DynamoDB, RDS, API Gateway, Step Functions), Google Vertex AI, Docker, Terraform, GitHub Actions, GitLab CI, Jenkins
- **Backend Frameworks:** Django, FastAPI, Flask
- **Data & Storage:** DynamoDB, PostgreSQL, Amazon RDS, Databricks, S3, Vector Databases (PG Vector)
- **Observability & Analytics:** CloudWatch, Structured Logging, Metrics & Alerting, GraphQL, Matplotlib, Hex
- **Data Storage & Retrieval:** DynamoDB, PostgreSQL, Amazon RDS, S3, pgvector (Vector Databases)
- **Engineering Practices:** Microservices Architecture, System Design, Performance Optimization, Agile Development

PUBLICATION

[A Novel Computational Framework of Robot Trust for Human-Robot Teams](#) - IEEE, 2025

- Developed a computational trust model enabling autonomous agents to quantify and dynamically update trust in human teammates.
- Implemented a multi-agent framework integrating psychological trust factors into adaptive robot decision-making.

[Interpretable Deep Learning for Biological Age Prediction: A Counterfactual Approach to Personalized Health Insights](#)

Atlantis Press, 2026

- Designed an Explainable Time Series Regression (XTSR) framework integrating deep learning with counterfactual reasoning.
- Trained hybrid time-series models on large-scale wearable sensor data to generate interpretable biological age predictions.

[Wildfire Prediction and Visualization: A Machine Learning Approach Using U.S. Data](#) - Atlantis Press, 2026

- Analyzed ~1.88 million U.S. wildfire incidents using machine learning and data visualization techniques.
- Identified patterns in frequency, distribution, and contributing causes using large-scale structured datasets.

CERTIFICATION

- **ISTQB Certified Tester (CTFL)** - ITB, Dec 2020
- **Build a Computer Vision App with Azure Cognitive Service** – Microsoft, Feb 2026